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Question 5 | Research Paper Summary

A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications

Research Purpose and Motivation

This paper examines how Retrieval-Augmented Generation (RAG) can improve the reliability of large language models in knowledge-intensive applications, especially finance and e-commerce. Although LLMs produce fluent answers, their internal knowledge can become outdated and they may invent financial figures, product details, or regulatory information. RAG reduces these problems by retrieving external evidence at the time of answering and supplying it to the model as context. The authors aim to establish which retrieval design offers the best balance of factual faithfulness, answer relevance, evidence coverage, traceability, and response latency. Rather than declaring a single architecture universally superior, the study evaluates the trade-offs needed for practical deployment.

Four RAG Architectures Compared

Sparse retrieval uses lexical matching, particularly BM25, to find documents sharing important terms with the query. Its rankings are fast and relatively interpretable, which is valuable for auditing exact financial or policy statements. However, sparse search can miss relevant documents when a user paraphrases a question, uses synonyms, or changes language.

Dense retrieval maps queries and documents into embedding vectors and searches for semantically similar passages using approximate nearest-neighbor methods. It handles paraphrasing and multilingual variation better than lexical search and increases semantic coverage. Its limitations include embedding and indexing costs and less transparent similarity scores.

Hybrid retrieval combines sparse and dense search, using methods such as reciprocal rank fusion or re-ranking to merge complementary results. It can preserve precise keyword matches while also recovering conceptually related evidence. This improves robustness, but requires managing two retrieval paths, duplicated passages, and greater system complexity.

Fusion retrieval uses multiple queries or retrieval stages, or combines evidence at the decoder level. Query expansion searches several related formulations of the user question, while fusion-in-decoder allows a model to integrate multiple passages during generation. This offers broad factual coverage and strong answer quality, but generally requires more computation, longer prompts, and more time; source attribution can also become harder.

Dataset and Experimental Design

The authors construct finance and e-commerce corpora from publicly accessible primary documents, including financial filings, official product information, and policy pages. Evaluation uses 500 questions stratified by domain and difficulty, with 75% answerable and 25% deliberately unanswerable cases. The zero-shot baseline receives no retrieved context, while RAG systems receive retrieved passages. To isolate retrieval effects, generator settings are fixed within comparison conditions. The paper discusses cloud-hosted GPT-4o and on-premises LLaMA deployments, with reproducible indexing configurations and telemetry for latency, sources, and token costs.

Evaluation Criteria and Analysis

The study adopts the RAGAS evaluation framework to assess faithfulness (whether claims are supported by evidence), answer relevancy, context precision (how useful retrieved passages are), and context recall (whether necessary evidence is retrieved). It additionally considers latency and audibility, defined around whether individual answer claims can be traced to cited passages. The protocol includes 95% bootstrap confidence intervals and paired statistical comparisons, rather than relying only on average scores. The authors also monitor retrieval parameters such as chunk sizes, top-k selection, embeddings, re-ranking, and indexing choices.

Main Results

All four retrieval-grounded approaches improve factual behavior over the zero-shot baseline. The paper reports that the ungrounded baseline invented monetary values or outdated regulatory thresholds in roughly one-third of the evaluated instances, whereas the RAG approaches reduced those errors to single-digit percentages. This demonstrates the importance of grounding generated claims in external documents rather than relying on model memory alone.

Sparse RAG achieves particularly strong context precision and audibility because exact lexical matches make evidence easy to inspect, but it suffers on paraphrased or multilingual questions. Dense RAG improves context recall and semantic coverage, although its rankings are less explainable. Hybrid RAG provides the most balanced overall results, combining lexical precision with semantic recall while maintaining practical response times. Fusion RAG attains the strongest faithfulness and answer relevance by collecting evidence from multiple angles, but introduces greater latency, token use, and complexity. Therefore, the “best” architecture depends on whether a system prioritizes speed, traceability, semantic breadth, or maximum factual coverage.

Limitations and Practical Implications

The evaluation is concentrated in finance and e-commerce, so results may not transfer unchanged to healthcare, education, or law. Performance also depends on the chosen generators, document collections, retrieval parameters, source freshness, and query characteristics. Automated RAGAS scores are useful proxies but do not replace detailed human review. The authors also acknowledge that comprehensive monetary cost and carbon-footprint accounting fall outside the study. Some production datasets and complete artifacts are restricted for contractual or data-protection reasons, limiting immediate independent replication.

For production systems requiring a clear audit trail and short latency, the study favors sparse or hybrid retrieval. For paraphrase-heavy or multilingual queries, dense or hybrid search offers better evidence discovery. Fusion is appropriate when factual completeness is more important than latency, for example in offline analysis or complex research, provided source attribution is carefully managed. Choosing authoritative, recent primary documents remains important regardless of architecture.

Conclusion

The paper concludes that RAG substantially strengthens factual reliability by connecting LLM answers to retrieved evidence. Sparse methods are transparent but narrow, dense methods are semantically flexible but harder to audit, hybrid methods deliver the most practical balance, and fusion methods maximize evidence coverage at higher operational cost. Its main contribution is a deployment-focused comparison that joins retrieval quality, answer faithfulness, auditability, and latency within one evaluation protocol. Future work proposed by the authors includes fresher dynamic indexing, adaptive retrieval and re-ranking, and closer integration between retrieval and generation.